Investigating Explainable Machine Learning Models for Dynamic Credit Scoring Using Behavioural Financial Data

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1. Introduction

Credit scoring is a fundamental mechanism used by financial institutions to evaluate borrower's creditworthiness, manage lending decisions, and manage credit risk (Rajender, et al., 2025). In Malaysia, credit assessment commonly relies on credit reporting agencies such as CCRIS and CTOS, which aggregate historical financing and repayment records from formal financial institutions (Bank Negara Malaysia, n.d.). While these systems have served as foundational tools for credit risk assessment, they are limited in their ability to reflect a borrower's financial evolves over time. At the same time, the rapid growth of digital financial services such as online banking, e-wallets and mobile payment platforms has generated large volumes of daily behavioural financial data that existing credit scoring systems do not systematically utilise. Furthermore, as machine learning models become increasingly adopted to improve credit risk prediction, many operate as opaque black-box systems that provide limited explanation regarding how credit decisions are generated. This raises significant ethical and regulatory concerns as every decision cannot be explained to regulators and borrowers. These three interconnected limitations, which are static data reliance, underutilised behavioural signals, and lack of model transparency represent a critical gap in contemporary credit scoring practice. This paper is organised as follows: Section 2 presents the literature review across three research domains and a similar systems comparison, Section 3 describes the research methodology, and Section 4 concludes.

2. Literature Review

2.1 Traditional Credit Scoring and Its Limitations

Credit scoring has served as the standard mechanism for evaluating borrower creditworthiness and managing lending risk across financial institutions. Conventional approaches apply standardised scoring models to generate a numerical risk rating that guides lending decisions using a structured credit bureau data such as repayment history, outstanding debt obligations, income stability, and employment status (Malik, Kaul, Gautham Jagthap, Harshit Jamley, & Chopra, 2025). These conventional models have been widely adopted due to their operational simplicity and regulatory acceptance to provide consistent and auditable assessments across large borrower populations (Malik, Kaul, Gautham Jagthap, Harshit Jamley, & Chopra, 2025). .

In Malaysia, credit assessment is supported by two principal credit reporting systems, which are CCRIS and CTOS. The Central Credit Reference Information System, or CCRIS is managed by Bank Negara Malaysia. CCRIS collects financing and repayment credit records submitted by financial institutions and is updated monthly. The data is then compiled by the 10th of each month and reflected in reports between the 10th and 15th (CTOS, 2026). On the other hand, CTOS is an approved credit reporting agency that uses CCRIS data with addition of legal records, trade references, and bankruptcy information to produce a more comprehensive credit profile. Together, these systems represent the foundational infrastructure of credit risk assessment in Malaysia and have demonstrated sustained utility in supporting structured lending decisions within the formal financial sector.

Although these systems remain useful, some limitations of traditional credit scoring have become increasingly noticeable as financial activities has shifted towards digital platforms, and the volume of alternative financial data has grown substantially. One of the limitations is traditional credit scoring mainly provides a historical view of a borrower's financial position. Even though CCRIS refreshes monthly, each update only reflects the same narrow categories of formal credit obligations such as loan repayments, credit card balances, and outstanding financing. This means that behavioural signals occurring between formal reporting cycles remain entirely invisible to the system (Malik, Kaul, Gautham Jagthap, Harshit Jamley, & Chopra, 2025). For example, a borrower may experience sudden reduction in income or may begin managing their finances more responsible through digital banking and e-wallet transaction. However, these changes may only be visible in the formal credit record after the later credit event, such as a delayed or missed repayment.

The second limitation is the systematic exclusion of thin-file borrowers. Thin-file borrowers are individuals who have little or no formal credit history. These individuals will appear invisible to credit bureaus because the system rely heavily on formal borrowing and repayment records. Therefore, even though they may actively use digital financial services and manage their finances responsibly, they may face difficulties when applying for a credit. The World Bank Group (2025) reported that approximately 1.4 billion adults worldwide remain financially underserved due to limited formal credit histories. This represents a structural exclusion embedded in traditional credit assessment models where current system disproportionately disadvantages economically active individuals in emerging digital economies. In Malaysia specifically, digital banking penetration exceeded 90 percent of the adult population by 2023. This means that the volume of behavioural financial data being generated through digital platforms has grown substantially while the structure of the formal credit reporting mechanisms remained unchanged (Bank Negara Malaysia).

The last limitation is the static architecture of the risk assessment. Traditional scoring models normally produce a risk score based on information available at a particular point in time, so the score may be slow to reflect changes in a borrower's financial circumstances, such as changes in income, spending patterns, repayment behaviour, or financial commitments (Verma & Kumar, 2024). As a result, financial institutions may be slower to identify increasing or decreasing borrower risk when those changes first appear through recent financial behaviour.

Overall, traditional credit scoring remains an important foundation for credit assessment because it provides structured and reliable historical information. However, its reliance on formal credit records and fixed-point assessments may limit its ability to reflect recent behavioural changes and assess thin-file borrowers. Therefore, complementary approaches are needed to analyse behavioural financial data and support more responsive, dynamic, and inclusive credit risk assessment.

2.2 Behavioural Financial Data and Machine Learning for Dynamic Credit Scoring

In today digital world, financial services such as online banking, e-wallets, mobile payment platforms, and buy-now-pay-later services is rapidly growing and has generated large volumes of daily behavioural financial data (Malik, Kaul, Gautham Jagthap, Harshit Jamley, & Chopra, 2025). Unlike formal credit records that only capture periodic information about existing credit obligations, behavioural data can reveal more recent patterns in how individuals spend, repay, transact, save, and manage their financial commitments. This pattern may provide more current and detailed view of a borrower's financial condition than bureau data alone can provide (Verma & Kumar, 2024; Malik, Kaul, Gautham Jagthap, Harshit Jamley, & Chopra, 2025).

Previous studies suggest that transactional and behavioural data can strengthen credit risk assessment by allowing changes in financial behaviour to be identified earlier. For example, Verma and Kumar (2024) demonstrated that incorporating behavioural transaction data into credit risk models improved classification accuracy by 15 to 18 percent compared to models relying fully on traditional bureau data, depending on the model configuration used.

Roland Abi (2025) further found that machine learning models trained on behavioural and transactional data can generate early warning signals up to 30 to 90 days before a borrower misses a formal payment, a capability structurally impossible for static bureau-based systems to replicate. This is particularly important because individual financial behaviour is not fixed but changes in response to income fluctuations, employment changes, spending habits, and personal circumstances. Therefore, a scoring approach that considers recent behavioural information may provide a more responsive assessment of changing borrower risk (Malik, Kaul, Gautham Jagthap, Harshit Jamley, & Chopra, 2025).

Based on the literature reviewed, four behavioural financial indicators have been consistently identified as measurable inputs for dynamic credit scoring, which are spending behaviour, repayment behaviour, transaction frequency, and financial management behaviour. Firstly, spending behaviour reflects how a borrower manages their outgoing financial resources. This includes whether stable, irregular, or excessive (Verma & Kumar, 2024). Excessive spending patterns may signal financial stress before a formal credit event such as a missed payment occurs, making it a useful early indicator of changing borrower risk. This makes it a useful early indicator of changing borrower risk Secondly, repayment behaviour. This indicator is one of the most direct because it reflects payment discipline and default risk(Rajender, et al., 2025; Abi, 2025). Research consistently identifies repayment consistency as among the strongest predictors of credit default, making it an essential input for any dynamic credit scoring framework. Thirdly, transaction frequency reflects borrower's level of financial engagement. This reflects a borrower's level of financial engagement and account activity(Malik, Kaul, Gautham Jagthap, Harshit Jamley, & Chopra, 2025). For example, regular transaction patterns can demonstrate active financial management, while irregular activity may reduce the predictability of a borrower's financial behaviour. Finally, the financial management behaviour. This captures how borrowers handle broader financial responsibilities including savings discipline, debt management, and budgeting consistency (Verma & Kumar, 2024; Jayanthi & Pruseth, 2026).

Overall, these behavioural indicators are broad categories rather than direct model inputs. Therefore, raw financial records must first be transformed through feature engineering into measurable variables that can be used as inputs for machine learning models. However, as behavioural financial data is large in volume, diverse, and continually evolving, the transformation process introduces analytical challenges that can be difficult to address using conventional statistical approaches alone (Verma & Kumar, 2024). Machine learning is therefore considered because it is suitable to identify complex non-linear relationships and detect hidden patterns within large and high-dimensional datasets (Jayanthi & Pruseth, 2026).

Three **machine learning** techniques have been identified in the literature as particularly suitable for analysing behavioural financial data in dynamic credit scoring, which are Random Forest, XGBoost, and Artificial Neural Networks (ANN). Random Forest is an ensemble method that handles non-linear relationships robustly and resists noisy behavioural data, characteristics important given the diversity of digital financial signals (Miracle, Shola, & Seun, 2024). Random Forest can also produce feature importance scores that provide initial insight into which behavioural indicators contribute most to predictions, offering a degree of interpretability not available in deep learning models (Abi, 2025). XGBoost has demonstrated consistently strong predictive performance on structured tabular financial data and is widely validated in credit risk prediction literature (Sampathkumar, Pathak, Ramaraj, Kotha, & Patel, 2026; Jayanthi & Pruseth, 2026). Lastly, Artificial Neural Networks can model highly complex non-linear relationships within large datasets, but their internal decision process is more difficult to interpret. (Abi, 2025; Malik, Kaul, Gautham Jagthap, Harshit Jamley, & Chopra, 2025)

Overall, behavioural financial data provides more recent information about borrower activity, while machine learning offers the analytical capability to identify complex patterns within that data. However, strong predictive capability alone is not sufficient for credit scoring because financial institutions and borrowers must also understand how a prediction is produced. Therefore, this creates the need for Explainable Artificial Intelligence.

2.3 Explainable Artificial Intelligence for Transparent Credit Scoring

Credit scoring decisions directly affect individuals' access to financial services such as loan approvals, credit limits, and interest rates. This makes transparency and accountability fundamental requirement of any automated credit assessment system rather than just a desirable feature (Izah, Ruiz, Olivier Gatete, Kumar, & Michael, 2025). In highly regulated domains like financial services, stakeholders such as regulators, financial institutions, and borrowers need to understand how automated predictions are generated rather than just know them as model that produces a number (Anang, et al., 2024).

When machine learning models operate as black-box systems, this requirement breaks down on two sides. Firstly, financial institutions face difficulties justifying automated credit decisions to regulators and customers. Secondly, borrowers who receive unfavourable credit decisions are unable to understand what specific financial behaviour contributed to the outcome or how they might improve their creditworthiness (Izah, Ruiz, Olivier Gatete, Kumar, & Michael, 2025; Rajender, et al., 2025). This can be further supported by research of Anang et al. (2024) that found 78 percent of fintech professionals surveyed considered model explainability essential for regulatory compliance. However, the same study also found that feature-level explanations remain limited in many existing credit scoring systems.

Moreover, the regulatory pressure at both global and local levels is increasing. In August 2024, the European Union AI Act formally classifies credit scoring as a high-risk AI application and requires transparent documentation, individual explanations, and human oversight for every decision made (Anang, et al., 2024). Furthermore, in Malaysia, Bank Negara Malaysia's responsible AI discussion paper (2022) and the Financial Sector Blueprint 2022 to 2026 both also signal the same direction, which is automated financial decisions must be explainable and auditable. This makes XAI not just academically relevant, but also necessary.

Explainable AI (XAI) refers to techniques that is designed to make the outputs of machine learning models interpretable and understandable to human users while preserving the predictive capabilities of the model (Ravi, Srivastava, Singh, Burila, & De, 2025). SHAP and LIME are techniques that have been widely identified in recent literature as particularly applicable to credit scoring contexts. SHapley Additive exPlanations (SHAP) is grounded in game theory and assigns each input feature a precise and mathematically justified contribution value for a specific prediction (Ravi, Srivastava, Singh, Burila, & De, 2025). SHAP works at two levels, which are local and global. For local, it can explain why a particular borrower received a particular score, while at global, it can show which specific features matter the most across all predictions (Rajender, et al., 2025). On the other hand, Local Interpretable Model-agnostic Explanations (LIME) explains specific predictions by constructing a simpler interpretable model built around that prediction to approximate how the complex ML model behaved for that case (Abi, 2025). Moreover, LIME is model-agnostic. This means that it can work with any of the ML models proposed in this framework without modification.

Despite the demonstrated capability of XAI techniques in recent studies, their integration into production credit scoring systems remains limited. Current fintech credit scoring implementations that is used by platform-based lenders and digital banks typically provide surface-level explanations only. For example, they present broad category labels such as "repayment history" or "credit utilisation" without quantifying how much each factor contributed to a specific decision or in which direction (Anang, et al., 2024;

Rajender, et al., 2025). This limits the practical utility of such explanations for both accountability and financial improvement, since borrowers cannot identify which behaviour to address or by how much. Collectively, these three domains reveal that traditional credit scoring, behavioural financial data, and explainable AI have each been examined individually, but limited research integrates all three within a unified framework, as the following similar systems analysis demonstrates.

2.4 Similar System

The comparison of five similar systems is presented in Table 1, where each system is evaluated based on its main approach, findings, and limitations relative to the present research.

Table 1 Similar system comparisons with research gaps identified

Journal / Conference (Year)	**Authors**	**Title**	**Key Findings**	**Research Gaps**
World Journal of Advanced Research and Reviews (2025)	Abi, R.	Machine Learning for Credit Scoring and Loan Default Prediction Using Behavioral and Transactional Financial Data	The integration of behavioural signals, such as spending patterns, payment timing, and mobile usage, with machine learning improves default prediction and may support financial inclusion in higher-risk environments.	Although the study discusses XAI techniques such as SHAP and LIME, it does not provide an integrated framework that addresses temporal data misalignment or dynamic feedback loops.
International Journal of Science, Engineering and Technology (2025)	Malik, P., Kaul, V., Jagthap, G., Jamley, H., and Chopra, H.	Dynamic Credit Scoring with Real-Time Transactional and Behavioral Data Using Deep Reinforcement Learning	The study formulates dynamic scoring as a Markov Decision Process and applies Deep Reinforcement Learning to update creditworthiness in real time, supporting continuous risk estimation and adaptive credit limit adjustment.	The study acknowledges the black-box nature of Deep Reinforcement Learning and identifies SHAP-based decomposition as a future direction, but it does not include a dedicated and validated XAI layer for real-time stakeholder explanations.
2025 7th International Conference on Information Systems and Computer Networks — ISCON (2025)	Rajender, C., Gururani, S., Mamadiyarov, Z., Toumasheva, M., Khorolskaya, T., and Naresh, R. B.	Explainable AI for Credit Scoring in Cloud Enterprise Banking Systems	A hybrid XAI approach combining LIME, SHAP, and rule-based aggregation achieved 89.2% accuracy and improved interpretability scores by 12% compared with traditional baseline models.	The framework was tested using a static historical dataset of 50,000 applications and does not address dynamic, high-frequency behavioural data or real-time score updates.
Journal of Advanced Computing Systems (2026)	Luo, Z., and Yu, M.	An Empirical Comparison of Feature Engineering Strategies from Non-Traditional Data for Thin-File Borrower Credit Assessment	Behavioural features derived from payment patterns produced the highest marginal predictive improvement for thin-file borrowers and reduced the performance gap compared with thick-file borrowers by 49.7%.	The study uses a static evaluation design and does not examine concept drift or how the importance of behavioural features may change over time as economic conditions evolve.
International Journal for Multidisciplinary Research (2026)	Jayanthi, M. K., and Pruseth, S.	An Explainable Machine Learning Framework for Dynamic Credit Portfolio Risk Assessment and Default Prediction Using	The study integrates an optimised Random Forest model with SHAP-based local and global explanations, reducing default misclassifi	The framework focuses mainly on portfolio-level risk and traditional financial indicators. It does not explicitly use high-

		Random Forest Optimisation	cation by 23% and supporting explainable credit assessment.	frequency behavioural transaction data as the main input for individual dynamic credit scores.

As summarised in Table 1, each system contributes to at least one element of behavioural data, dynamic scoring, machine learning, or explainability, but none integrate all four within a single framework. Furthermore, several rely on static datasets that do not support continuous behavioural updates.

3. Methodology

3.1 Target user

The proposed explainable dynamic credit scoring framework is designed to serve three primary user groups within the financial services ecosystem.

The first user group is financial institutions, including commercial banks, digital banks, and fintech lenders, whose risk management teams and credit officers would use the framework to generate dynamic credit score predictions based on borrower behavioural financial data.

The second user group is borrowers. This includes individuals who apply for credit products including personal loans, credit cards, and buy-now-pay-later facilities. Borrowers may benefit from the framework through XAI-generated explanations that accompany credit decisions. This feature enables them to understand which specific financial behaviours contributed positively or negatively to their predicted credit score and what corrective actions are available to improve their creditworthiness over time.

The third user group is credit analysts and professionals that responsible for interpreting model outputs and supporting lending decisions. This group can benefit from the decision and behavioural explanation provided by the framework. Therefore, the lending decisions will be more transparent and auditable than pure black-box ML systems currently exist.

3.2 Sampling method

This study uses stratified sampling as the primary sampling strategy. This sampling type is selected because the target respondents consist of two distinct groups, which are financial service users and financial or fintech professionals. Both groups are important because they. Users represent borrower perspective, while professionals provide institutional and practical insights. Moreover, a total target sample of 150 respondents is proposed, which consists of 120 financial service users and 30 financial or fintech professionals. Convenience sampling will then be applied within each group to ensure more practical data collection.

For financial service user group, participants must age 18 or above and are actively using digital financial services such as online banking, e-wallets or credit card. Participants will be recruited through online channels such as social media platforms and digital financial service communities.

Next, for the financial or fintech professional group, participants must have direct experience in credit assessment and financial operation. For example, banking staff, credit officers, fintech employees, data analysts, and risk management professionals. Professional participants will be recruited through professional networks such as LinkedIn and academic contacts.

3.3 Data collection method

Quantitative research approach is used with a structured questionnaire. This approach is suitable because it allows measurable information to be collected from large number of respondents at a time. Questionnaire items will focus on behavioural financial indicators, dynamic credit scoring, and perceptions towards transparency in automated credit decision.

The questionnaire will be distributed via an online survey platform and be divided into four sections. The first section collects demographic information such as age, employment status, and experience with digital financial services. The second section will collect respondent's view on what behavioural financial factors related to them. For example, spending behaviour, repayment behaviour, transaction frequency, and financial management behaviour. The third section examine respondent's acceptance of dynamic credit scoring, such as whether credit scores should be updated when financial behaviour changes. And the last section collects respondent's opinion on explainable AI. For instance, whether borrower should receive clear explanation about the factor that influence their credit score. For professionals, additional items will also be included such as their stand on suitable behavioural data in credit assessment and practical use of ML and XAI.

4. Conclusion

In conclusion, this study focused on two main objectives. Firstly, it examined the limitations of traditional credit scoring systems that mainly rely on historical credit records and static risk assessments. Secondly, it proposed an integrated conceptual framework that combines behavioural financial data, machine learning models, and explainable artificial intelligence to support dynamic credit scoring. Previous research shows that borrower financial behaviour can change over time due to income fluctuations, spending patterns, repayment habits, transaction activity, and financial management practices. However, these changes may not be reflected immediately in traditional credit records because existing credit scoring systems mainly depend on formal information such as loan repayments, outstanding debt, and credit history. As a result, financial institutions may not be able to identify changes in borrower risk at an early stage, while thin-file borrowers may also receive limited assessment due to the lack of formal credit records. Through the proposed framework, four behavioural indicators are included in the credit assessment process, namely spending behaviour, repayment behaviour, transaction frequency, and financial management behaviour. These indicators are analysed using machine learning models to generate updated credit risk predictions, while XAI techniques such as SHAP and LIME are used to explain how each behavioural factor influences the prediction. The comparison of similar systems further shows that although behavioural data, dynamic scoring, machine learning, and explainability have been explored in previous studies, limited research integrates all of these components within one framework for explainable dynamic credit scoring. By establishing the conceptual foundation for this framework, this study provides a useful basis for further empirical research, model evaluation, framework refinement, and the development of more transparent and adaptive credit assessment system.

5. Appendix

DRAFT QUESTIONNAIRE

Explainable Dynamic Credit Scoring Using Behavioural Financial Data

Section A: Screening and Demographic Information

Respondent Group	☐ Financial service user / borrower ☐ Financial or fintech professional
Age	☐ 18–25 ☐ 26–35 ☐ 36–45 ☐ 46 and above
Employment Status	☐ Student ☐ Employed ☐ Self-employed ☐ Unemployed ☐ Other: ________
Digital Services Used	☐ Online banking ☐ E-wallet ☐ Credit card ☐ BNPL ☐ Mobile payment ☐ Other: ________
Credit Experience	☐ Personal loan ☐ Credit card ☐ BNPL ☐ None ☐ Other: ________
Professional Area (if applicable)	☐ Banking ☐ Credit assessment ☐ Risk management ☐ Fintech ☐ Data analytics ☐ Other: ________

Section B: Spending Behaviour

Please indicate your level of agreement: 1 = Strongly Disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly Agree

Code	Statement	1	2	3	4	5
SB1	Spending stability should be considered when assessing borrower creditworthiness.	☐	☐	☐	☐	☐
SB2	Sudden increases in spending may indicate a change in a borrower's financial risk.	☐	☐	☐	☐	☐
SB3	Spending that is consistently high compared with income may indicate financial pressure.	☐	☐	☐	☐	☐
SB4	The balance between essential and non-essential spending can provide useful credit-risk information.	☐	☐	☐	☐	☐
SB5	Recent spending patterns should complement formal credit history in credit assessment.	☐	☐	☐	☐	☐

Section C: Repayment Behaviour

Please indicate your level of agreement: 1 = Strongly Disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly Agree

Code	Statement	1	2	3	4	5
RB1	Consistent on-time repayments indicate stronger payment discipline.	☐	☐	☐	☐	☐
RB2	Repeated late or missed payments should increase a borrower's assessed credit risk.	☐	☐	☐	☐	☐
RB3	Payment records for loans, bills, and other commitments can provide useful behavioural information.	☐	☐	☐	☐	☐
RB4	Recent repayment behaviour should influence an updated credit-risk assessment.	☐	☐	☐	☐	☐
RB5	Repayment consistency is one of the most important indicators of borrower reliability.	☐	☐	☐	☐	☐

Section D: Transaction Frequency

Please indicate your level of agreement: 1 = Strongly Disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly Agree

Code	Statement	1	2	3	4	5
TF1	Regular account activity can provide useful information about a borrower's financial engagement.	☐	☐	☐	☐	☐
TF2	Large changes in transaction frequency may indicate a change in financial circumstances.	☐	☐	☐	☐	☐
TF3	Deposit and withdrawal frequency can help describe a borrower's cash-flow pattern.	☐	☐	☐	☐	☐
TF4	Digital transaction records may support the assessment of borrowers with limited formal credit history.	☐	☐	☐	☐	☐
TF5	Transaction regularity should be considered together with other behavioural indicators.	☐	☐	☐	☐	☐

Section E: Financial Management Behaviour

Please indicate your level of agreement: 1 = Strongly Disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly Agree

Code	Statement	1	2	3	4	5
FM1	Regular saving behaviour can indicate responsible financial management.	☐	☐	☐	☐	☐
FM2	Budgeting consistency should be considered when assessing financial responsibility.	☐	☐	☐	☐	☐
FM3	The ability to manage several financial commitments may reflect borrower reliability.	☐	☐	☐	☐	☐
FM4	Changes in account balance and savings patterns can provide useful credit-risk information.	☐	☐	☐	☐	☐
FM5	Overall financial management behaviour should complement formal credit records.	☐	☐	☐	☐	☐

Section F: Dynamic Credit Scoring

Please indicate your level of agreement: 1 = Strongly Disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly Agree

Code	Statement	1	2	3	4	5
DCS1	Credit-risk assessments should be updated when a borrower's financial behaviour changes.	☐	☐	☐	☐	☐
DCS2	Using recent behavioural data can provide a more responsive assessment than relying only on historical records.	☐	☐	☐	☐	☐
DCS3	Dynamic credit scoring may help financial institutions identify changes in borrower risk earlier.	☐	☐	☐	☐	☐
DCS4	Improved financial behaviour should be reflected in later credit-risk assessments.	☐	☐	☐	☐	☐
DCS5	New behavioural information should be considered without requiring the model to be retrained after every transaction.	☐	☐	☐	☐	☐

Section G: Explainable AI and Transparency

Please indicate your level of agreement: 1 = Strongly Disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly Agree

Code	Statement	1	2	3	4	5
XAI1	Borrowers should receive clear explanations for automated credit decisions.	☐	☐	☐	☐	☐
XAI2	An explanation should show which behavioural factors increased or reduced the predicted credit risk.	☐	☐	☐	☐	☐
XAI3	Financial institutions should be able to justify AI-assisted credit decisions to customers and regulators.	☐	☐	☐	☐	☐
XAI4	I would trust an AI-assisted credit assessment more if its decision could be explained clearly.	☐	☐	☐	☐	☐
XAI5	Explainability is as important as predictive performance in credit scoring.	☐	☐	☐	☐	☐

Section H: Professional Perspective (Professionals Only)

Please indicate your level of agreement: 1 = Strongly Disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly Agree

Code	Statement	1	2	3	4	5
PP1	Behavioural financial data is practical for supporting credit assessment within financial institutions.	☐	☐	☐	☐	☐
PP2	Random Forest, XGBoost, and Artificial Neural Networks are relevant models to investigate for behavioural credit scoring.	☐	☐	☐	☐	☐
PP3	SHAP and LIME are relevant techniques for explaining machine-learning credit predictions.	☐	☐	☐	☐	☐
PP4	Data privacy, model governance, and regulatory compliance are major concerns in implementing dynamic credit scoring.	☐	☐	☐	☐	☐
PP5	An integrated framework combining behavioural data, machine learning, and XAI would support more transparent lending decisions.	☐	☐	☐	☐	☐

6. References

Abi, N. R. (10 June, 2025). Machine learning for credit scoring and loan default prediction using behavioral and transactional financial data. *World Journal of Advanced Research and Reviews, 26*(3), 884-904. doi:10.30574/wjarr.2025.26.3.2266

Anang, A., Ajewumi, O., Sonubi, T., Chukwujekwu, K., Arogundade, J., & Akinbi, I. (30 October, 2024). Explainable AI in financial technologies: Balancing innovation with regulatory compliance. *International Journal of Science and Research Archive*(1), 1793-1806. doi:10.30574/ijsra.2024.13.1.1870

Bank Negara Malaysia. (n.d.). *CCRIS Report - Bank Negara Malaysia*. Retrieved from www.bnm.gov.my

CTOS. (9 April, 2026). *CTOS vs ccris: What's the difference and why it matters in Malaysia*. Retrieved from CTOS - malaysia's leading credit reporting agency: https://ctoscredit.com.my/learn/ctos-vs-ccris-whats-the-difference-and-why-it-matters-in-malaysia/?srsltid=afmboopiuamvlo5h02-6opr0xhaegq59ntiu0elsjz5ripadohyac5o1

Izah, C., Ruiz, A., Olivier Gatete, Kumar, D., & Michael, I. (25 September, 2025). The Role of Machine Learning in Enhancing Credit Scoring Models for Financial Inclusion. Retrieved from https://www.researchgate.net/publication/395822244_The_Role_of_Machine_Learning_in_Enhancing_Credit_Scoring_Models_for_Financial_Inclusion

Jayanthi, D., & Pruseth, S. (April, 2026). An Explainable Machine Learning Framework for Dynamic Credit Portfolio Risk Assessment and Default Prediction Using Random Forest Optimization. *6*(2). Retrieved from https://www.ijfmr.com/papers/2026/2/73602.pdf

Luo, Z., & Yu, M. (11 April, 2026). An empirical comparison of Feature engineering strategies from Non-Traditional data for Thin-File borrower credit assessment. *Journal of Advanced computing systems, 6*(4), 50-59. doi:10.69987/jacs.2026.60404

Malik, P., Kaul, V., Gautham Jagthap, Harshit Jamley, & Chopra, H. (11 November, 2025). Dynamic Credit Scoring with Real-Time Transactional & Behavioral Data Using Deep Reinforcement Learning. *RIET-IJSET International Journal of Science Engineering and Technology, 13*(5), 2395-4752. Retrieved from https://www.researchgate.net/publication/397482338_Dynamic_Credit_Scoring_with_Real-Time_Transactional_Behavioral_Data

_Using_Deep_Reinforcement_Learning

Miracle, A., Shola, A., & Seun, O. (1 December, 2024). THE IMPACT OF MACHINE LEARNING ON CREDIT RISK MODELLING. Retrieved from https://www.researchgate.net/publication/386737866_THE_IMPACT_OF_MACHINE_LEARNING_ON_CREDIT_RISK_MODELLING

Rajender, C., Zokir, M., Tatyana, K., Samriddhi, G., Marina, T., & Rohith Bommalla, N. (5 September, 2025). Explainable AI for Credit Scoring in Cloud Enterprise Banking Systems. *2025 7th International Conference on Information Systems and Computer Networks (ISCON)*, 1-6. doi:10.1109/iscon65210.2025.11341493

Ravi, V., Srivastava, V., Singh, M., Burila, R., & De, I. (14 March, 2025). Explainable AI (XAI) for Credit Scoring and Loan Approvals. Retrieved from https://www.researchgate.net/publication/389847187_Explainable_AI_XAI_for_Credit_Scoring_and_Loan_Approvals

Sampathkumar, V., Pathak, I., Ramaraj, D., Kotha, R., & Patel, D. (5 January, 2026). Federated Explainable AI for Fair and Inclusive Credit Scoring: A Comprehensive Review. *2026 IEEE 16th Annual Computing and Communication Workshop and Conference (CCWC)*, 0142-0151. doi:10.1109/ccwc67433.2026.11393798

Verma, P., & Kumar, K. (2 December, 2024). Behavioral Analytics for Credit Scoring: Beyond Traditional Metrics. Retrieved from https://www.researchgate.net/publication/394086476_Behavioral_Analytics_for_Credit_Scoring_Beyond_Traditional_Metrics

World Bank Group. (2025). *Financial Inclusion | World Bank Group*. Retrieved from Worldbank.org: https://www.worldbank.org/ext/en/topic/financial-sector/financial-inclusion